JobMatch AI Chrome Extension

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Objective

Job searching has become increasingly time-consuming as AI-generated postings and high-volume listings make it harder to quickly assess fit. While LinkedIn offers a built-in skill-matching feature, most major job boards, such as Workday, do not. This project fills that gap by providing a Chrome extension that brings LinkedIn-style skill matching to multiple platforms.

Architecture

The system is built on a lightweight four-layer architecture (see Figure 1). At the top is the **User layer**, where job seekers browse postings across supported job boards. The **Extension layer** powers the in-browser experience: it extracts job descriptions from the page, provides the side-panel interface, and coordinates communication with the backend. The **Backend layer** consists of a Cloudflare Worker that routes requests, enforces rate limits, and securely proxies calls to the AI model. At the core, the **AI layer** uses Groq's Llama 3.3 70B model to analyze the resume and job description, generate match scores, identify skill gaps, and produce ATS-optimized resume content.

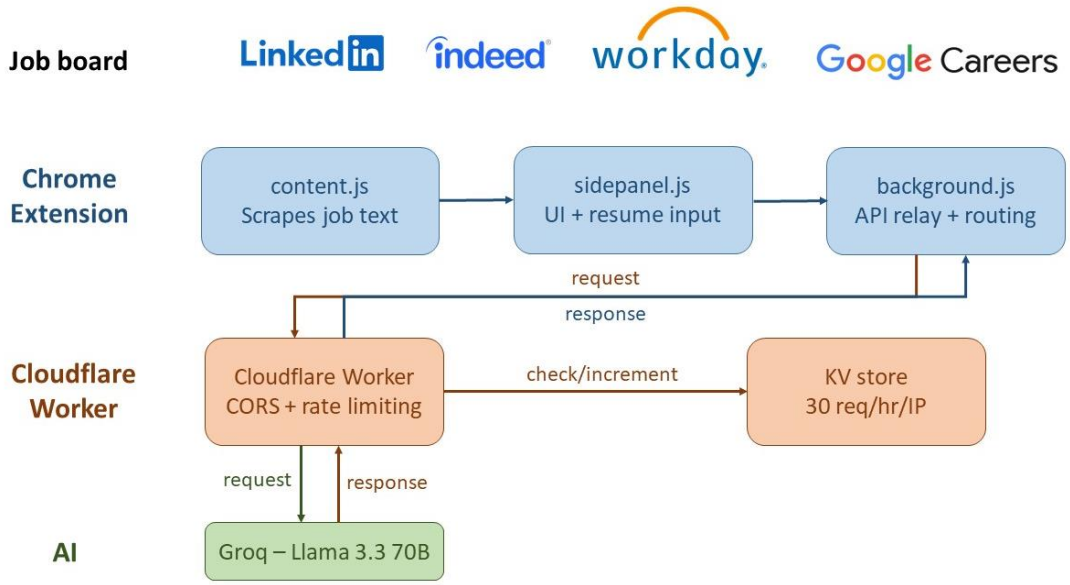

Figure 1. Architecture Overview.
The system consists of four layers: user/job board, Chrome extension, Cloudflare Worker backend, and Groq-powered AI processing.

Key Features

JobMatch AI provides a streamlined interface with three core features (see Figure 2). The **Match** tab analyzes a user's resume against any job description, offering an instant match score and skill breakdown, with automatic job-description scraping on supported platforms. The **ATS Writer** generates optimized resume sections — such as summaries, bullet points, and skills — using the user's saved resume and the current job posting, with multiple tone options for different application styles. The **History** tab stores past analyses, allowing users to revisit results and compare how their fit varies across roles over time.

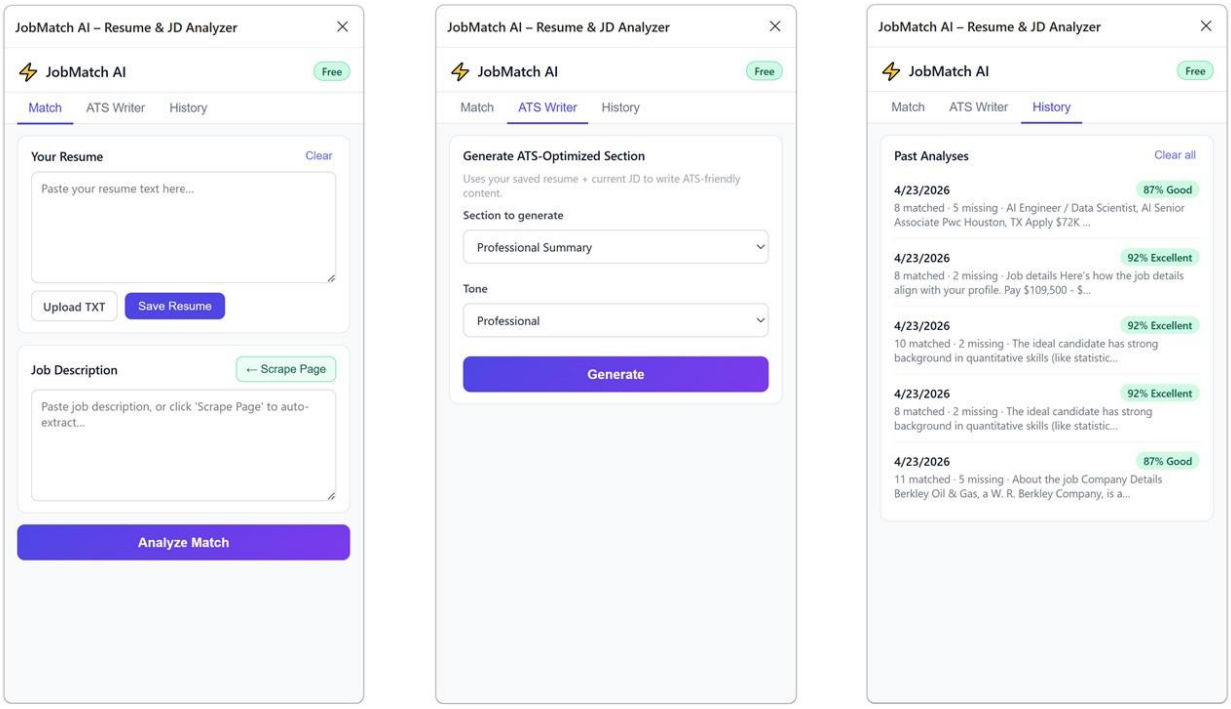

Figure 2. The UI has three tabs: Match (left) for resume–JD analysis, ATS Writer (middle) for generating optimized sections, and History (right) for past results.

Conclusion

JobMatch AI closes a critical gap in today's job-search ecosystem by bringing instant skill matching and ATS-optimized rewriting to platforms that lack these capabilities. The project highlights the power of combining a Chrome extension, a serverless backend, and Groq-accelerated LLMs to deliver real-time insights with minimal cost. With its modular design, the system is well-positioned for future expansion, including broader platform coverage and more advanced resume-generation features.